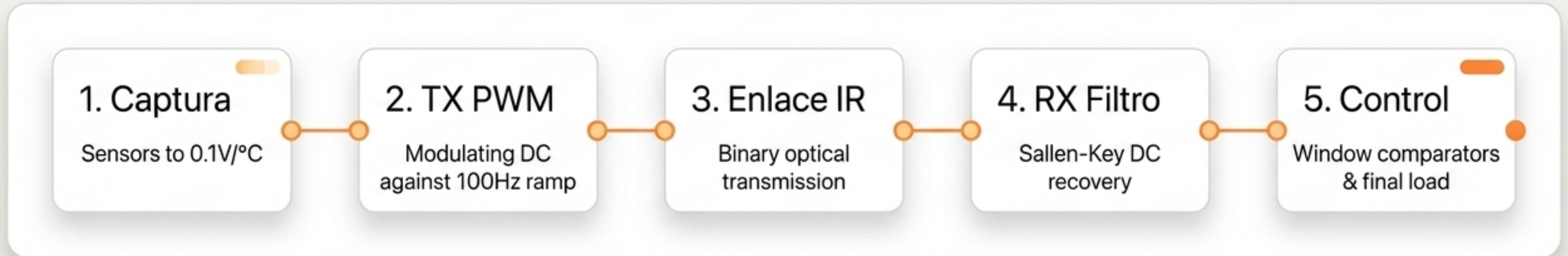


Industrial Telemetry & PWM-IR Control

Systems Architecture
& Signal Journey



The Macro Architecture



The Objective

Process temperature from two points, average them, and transmit the data via optical telemetry (Infrared) using Pulse Width Modulation (PWM) to control an extractor fan.

The Challenge

'Ruido Óptico' — external light interference physically degrades data quality across the unshielded air gap.

Stage 1: Capture & Translation

RTD (Pt100)

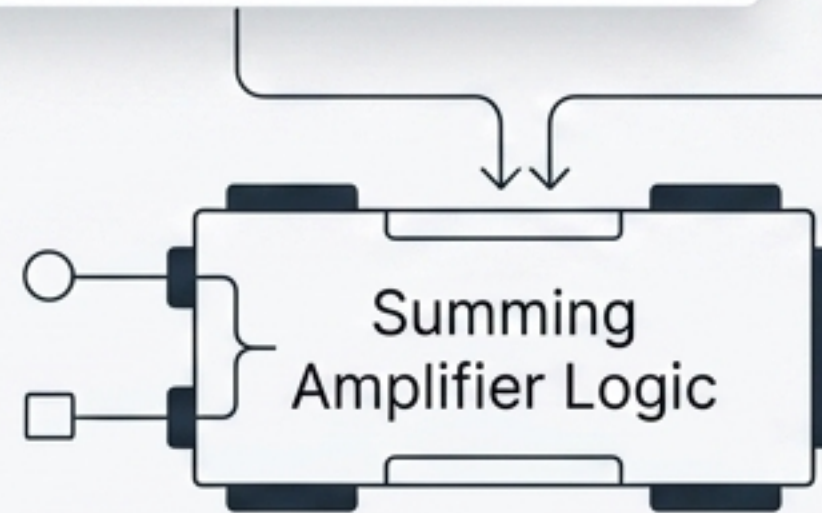
Industrial standard, high precision, resistive.

25.0 °C

Analog Sensor (LM35)

IC-based, linear voltage output, local board monitoring.

30.0 °C

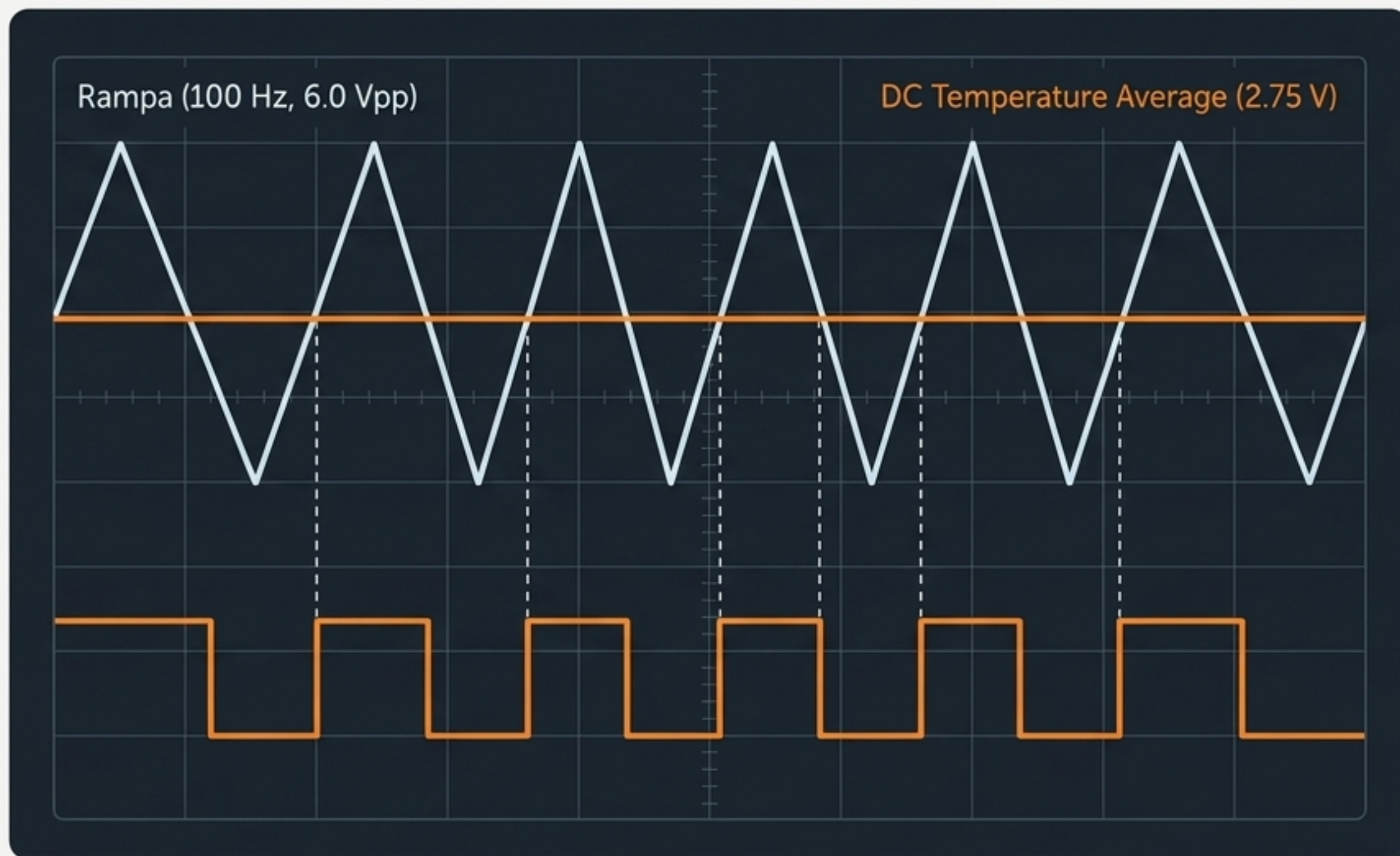


$25.0\text{ °C} + 30.0\text{ °C} \rightarrow \text{Average: } 27.5\text{ °C}$

Continuous DC Baseline: 2.75 V

Scale: 0.1V/°C

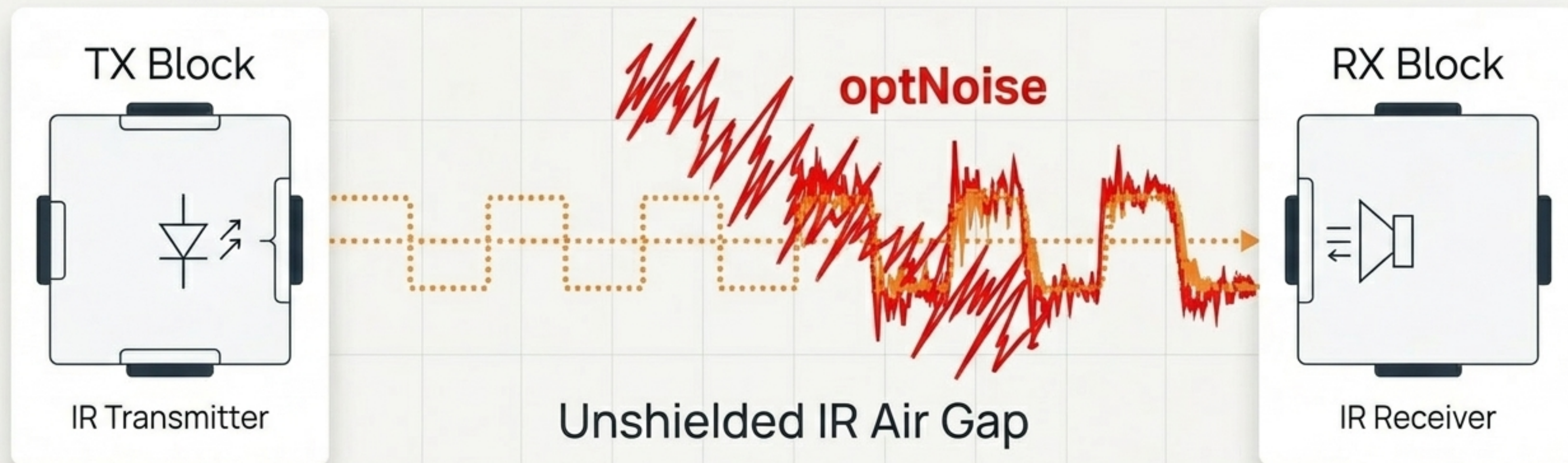
Stage 2: The **PWM** Intersection Engine



The Mechanism: Op-Amp comparators match the flat temperature voltage against a high-frequency ramp, encoding analog amplitude into the width of a digital pulse.

The Advantage: Digital pulses are highly resistant to amplitude degradation over physical distances.

Stage 3: The Void & The Noise



Electricity Becomes Photons: The PWM square wave fires an IR LED, transmitting data optically.

The Vulnerability: Ambient light, factory strobes, or sunlight violently distort the physical amplitude of the arriving light.

Stage 4: Sallen-Key Signal Demodulation



Signal Recovery

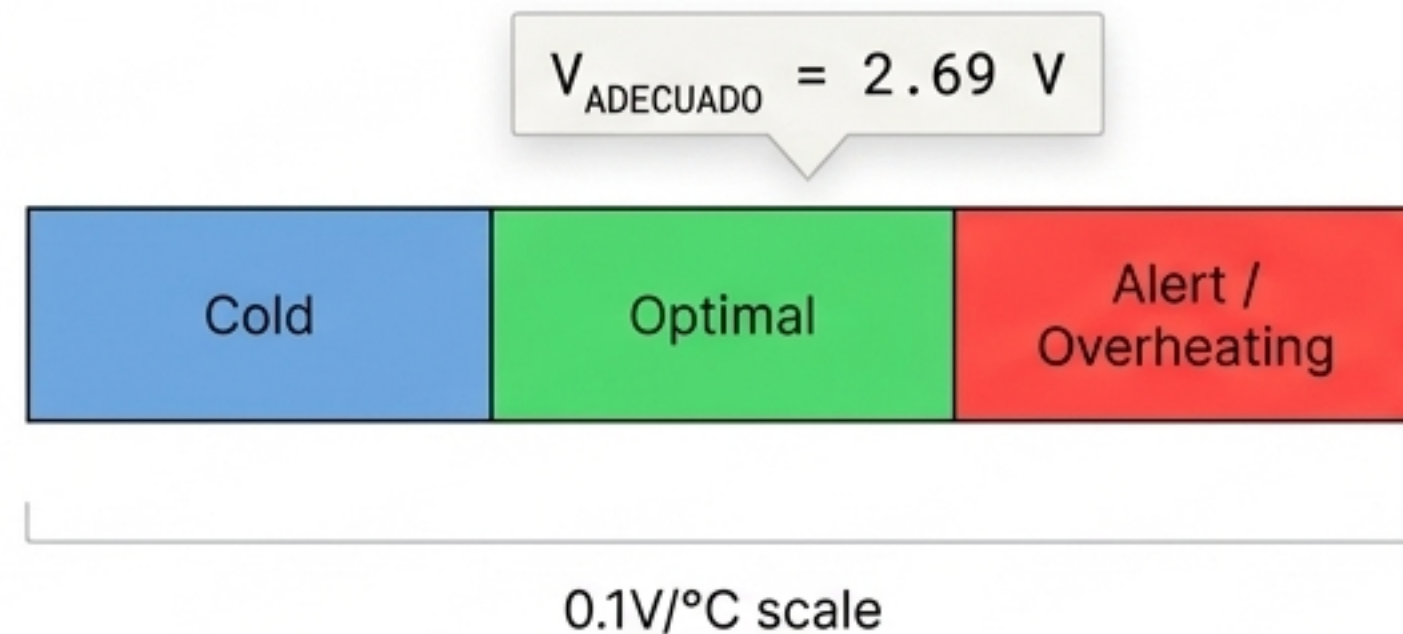
Because the temperature data is encoded in the timing (width) of the pulses, not their height, the physical amplitude noise is rendered irrelevant.

The Math

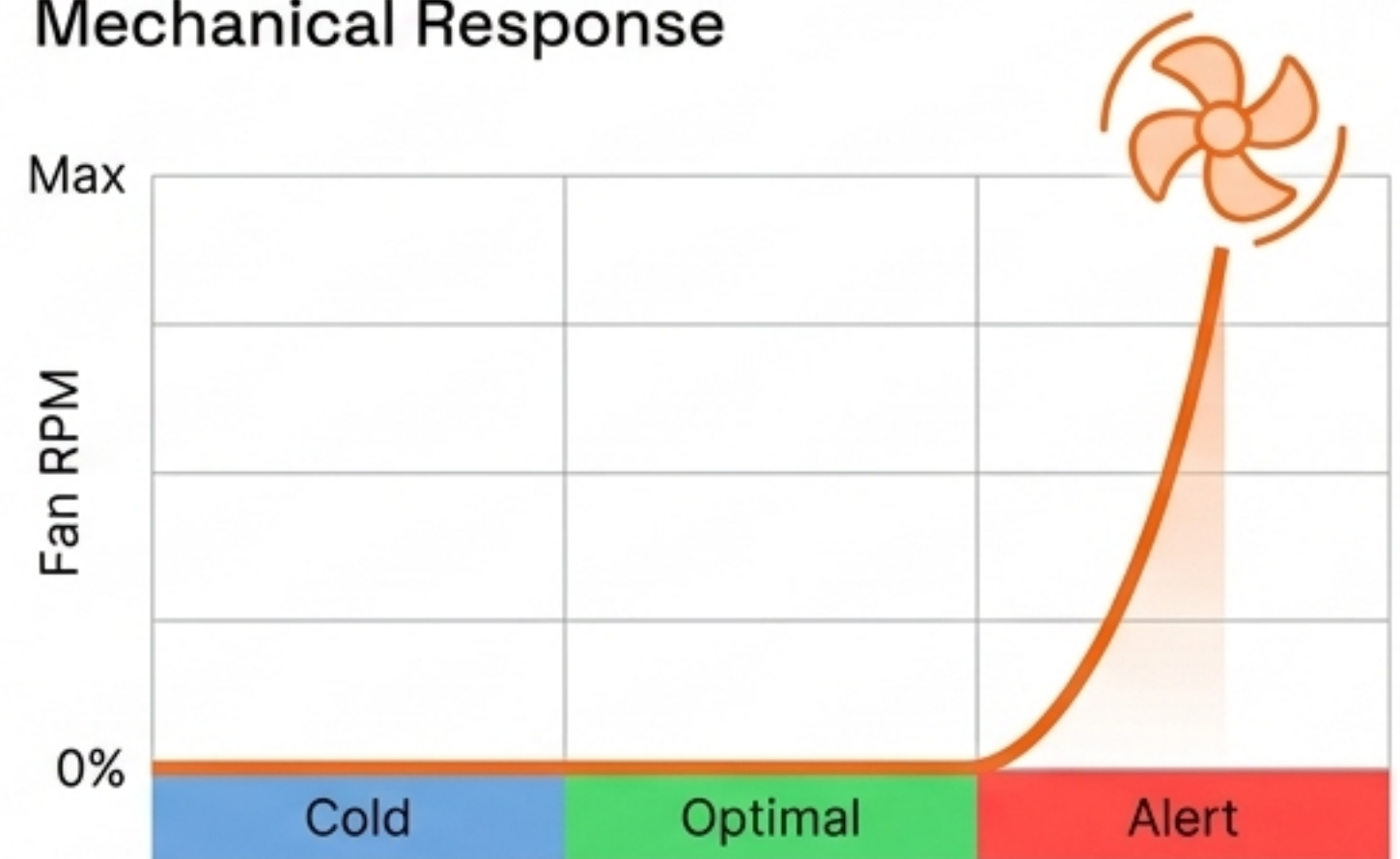
The analog Sallen-Key low-pass filter averages out both the high-frequency PWM carrier and the optical noise, perfectly reconstructing the original continuous DC voltage.

Stage 5: Mechanical Control Logic

Window Comparator Gauge



Mechanical Response



Decision Logic: Window comparators translate the recovered analog voltage back into targeted, automated mechanical action, defining strict operational boundaries for the cooling system.

Signal State Diagnostic Matrix

Node A (Sensors)



Continuous DC.
Variable by
temperature.

Highly precise, but
highly vulnerable to
degradation over long
physical distances.

Node B (TX Output)



Square Wave (PWM).
100Hz Carrier.

Data paradigm shift.
Analog amplitude is
now encoded entirely
in pulse timing.

Node C (Air Gap)



Optical Pulses
(Infrared). Polluted
by optNoise.

Amplitude is violently
corrupted by light
interference, but
frequency timing
remains intact.

Node D (RX Output)

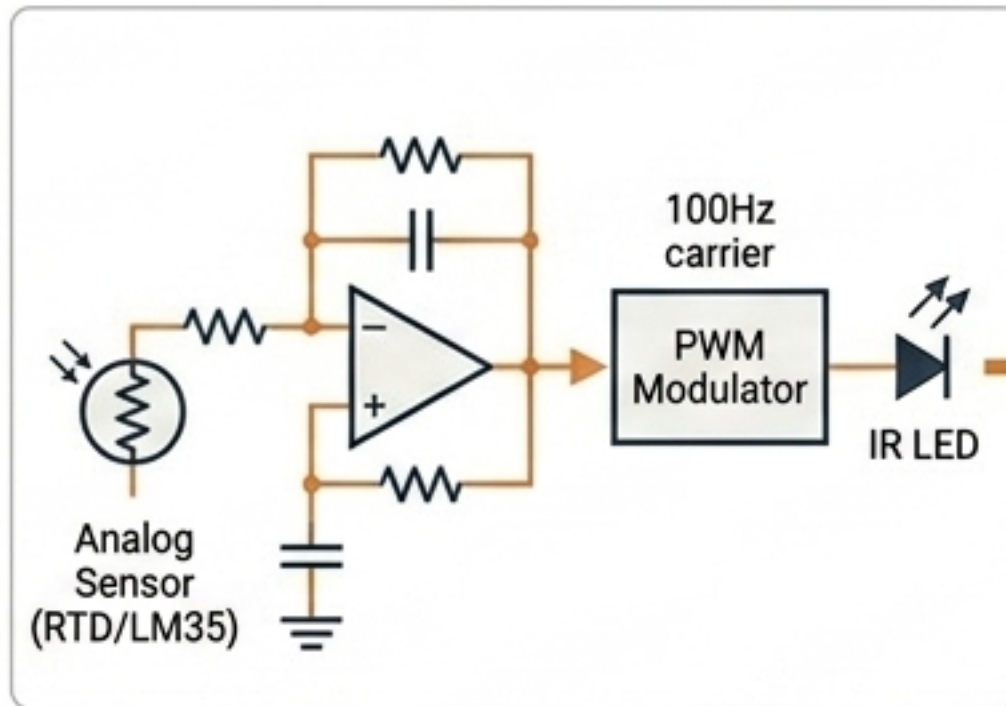


Recovered
Continuous DC.

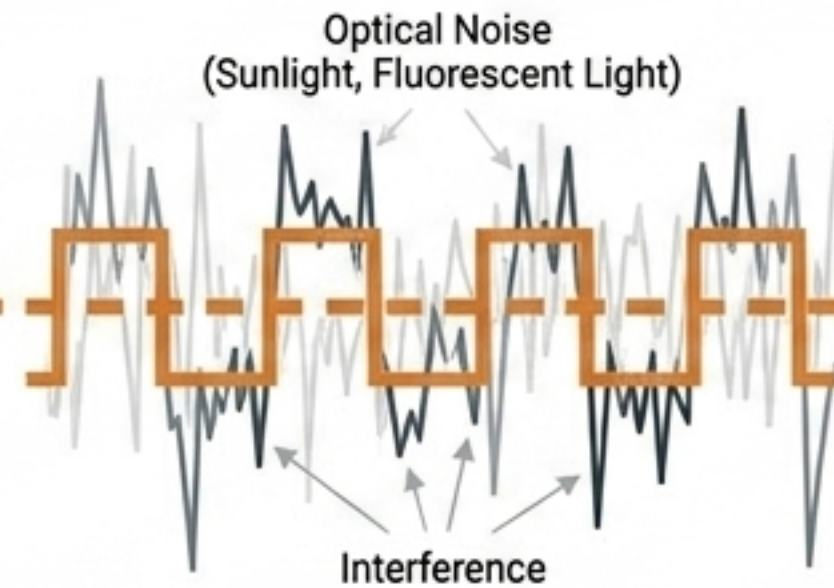
Noise stripped away
by Sallen-Key filtering.
Pure analog data ready
for mechanical logic.

The Anatomy of Signal Resilience

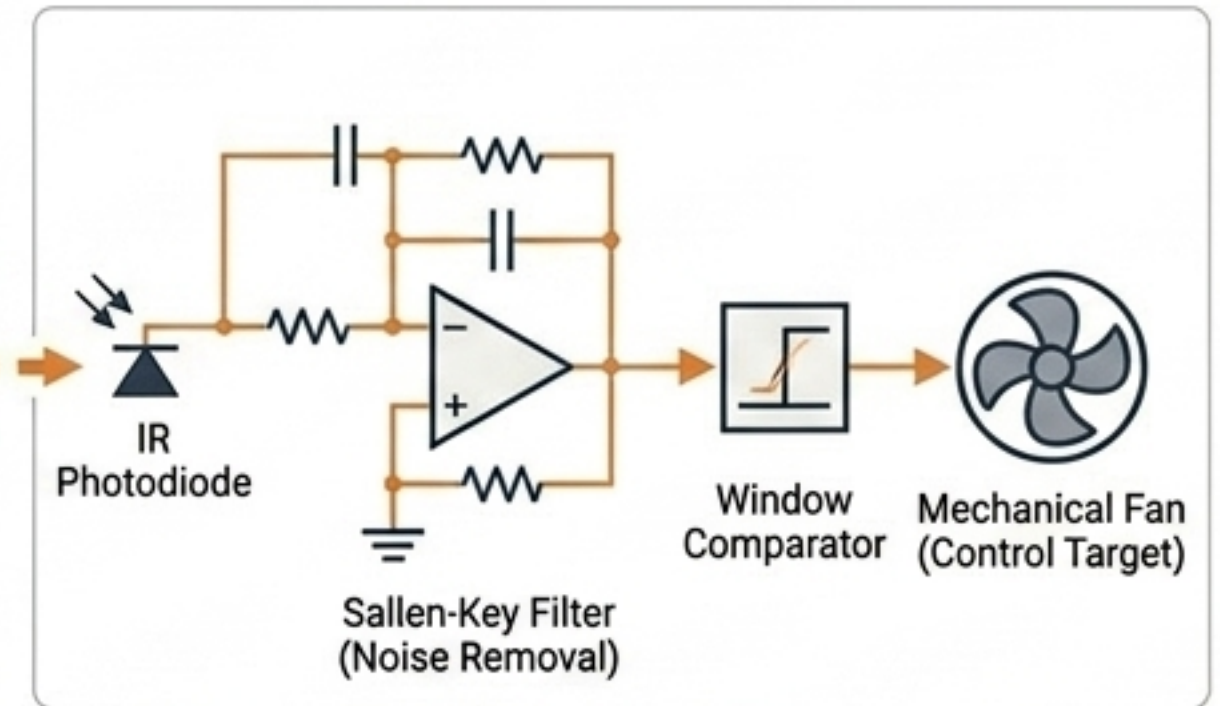
Zone 1: Secure TX Environment



Zone 2: The Chaotic Void (Unshielded air gap)



Zone 3: Secure RX Environment



The Core Engineering Insight

Why use PWM and IR instead of simply sending a raw analog voltage via a long wire?

Raw voltage degrades linearly over distance and is permanently corrupted by electromagnetic interference. By converting to PWM, the architecture offloads data integrity onto *time* rather than *amplitude*. Optical noise degrades light intensity, but it cannot alter the width of the pulse. The system mathematically guarantees its own integrity.

Engineering Contextualized: **The AI Tutor Layer**

PREGUNTA A RIGOBERTO

What is the princess and reintvat to the input at trupktar?



Hello, us really since their consectative edpils underanting the system concept.

Omg.., "Why did || you retorse messenger is oscilloscope: towi as its aralog," or well, it my Sallen-Key filter. the voltage is use the siow and use reveran interfereps in voltage.

Tu duda técnica...



Rigoberto

An embedded AI Technical Tutor powered by the Groq API.

Contextual Interrogation

Students can pause the real-time oscilloscope and directly interrogate the system's logic (e.g., 'Why did the Sallen-Key filter drop the voltage by 0.06V?').

Dynamic Learning

Transforms a static diagram and raw telemetry data into a queryable, interactive educational environment.